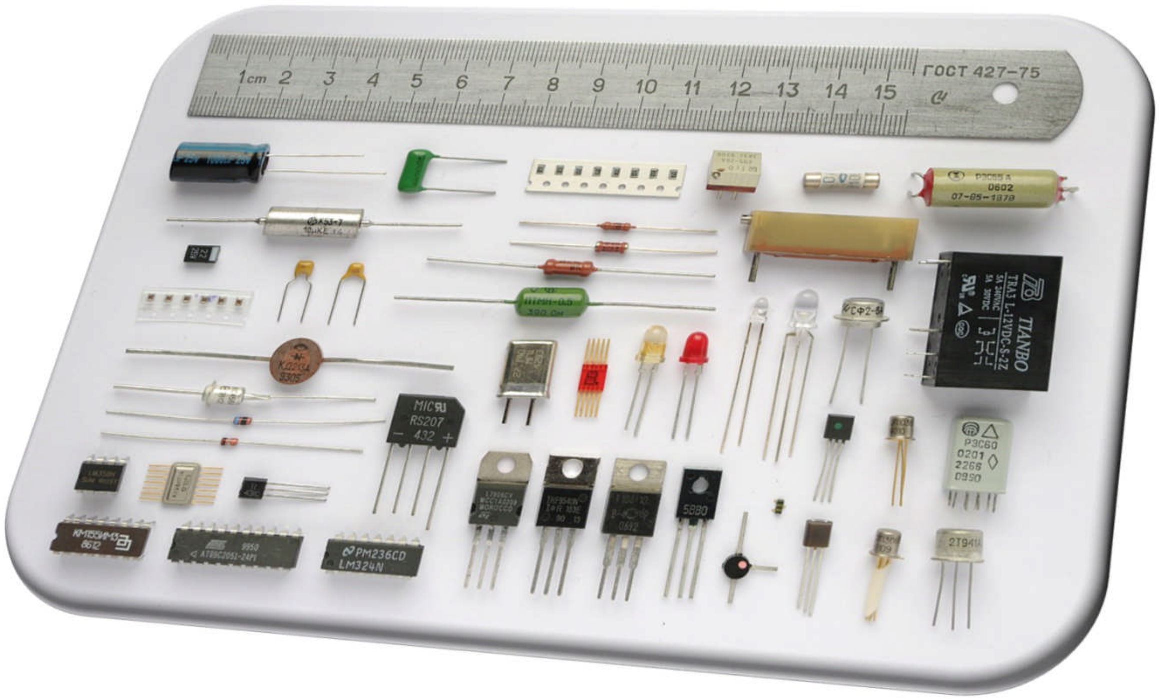
Unit 2
Electronics Concept

Objective

- Electronics components
- Logic Gates – AND, OR, NOR, NAND, NOT
- Electronics Signals
- Pulse width Modulation
- ADC – Analog to Digital Converter

Electronics Components

- An electronic component is a physical entity in an electronic system used to affect movement of electrons. Electronic components have a number of electronic terminals or leads which connect to other electronic components over wire to create an electronic circuit.



https://en.wikipedia.org/wiki/File:Componentes.JPG

Electronics Components

Electronics Components

Active Components

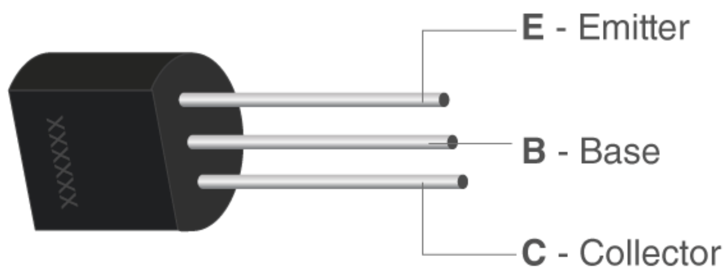
Transistors, Amplifiers

Passive Components

Register, inductor, Capacitor

Transistor

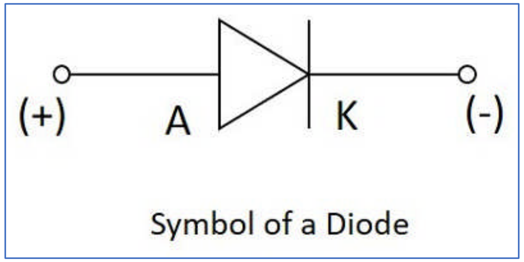
- A transistor is a semiconductor device used to amplify and switch electronic signals and electrical power.
- Application of Transistor
  - Power Regulator Circuits
  - Microprocessor ICs
  - Mobile Phone charger
  - AC to DC adaptors
  - Electronics switching devices

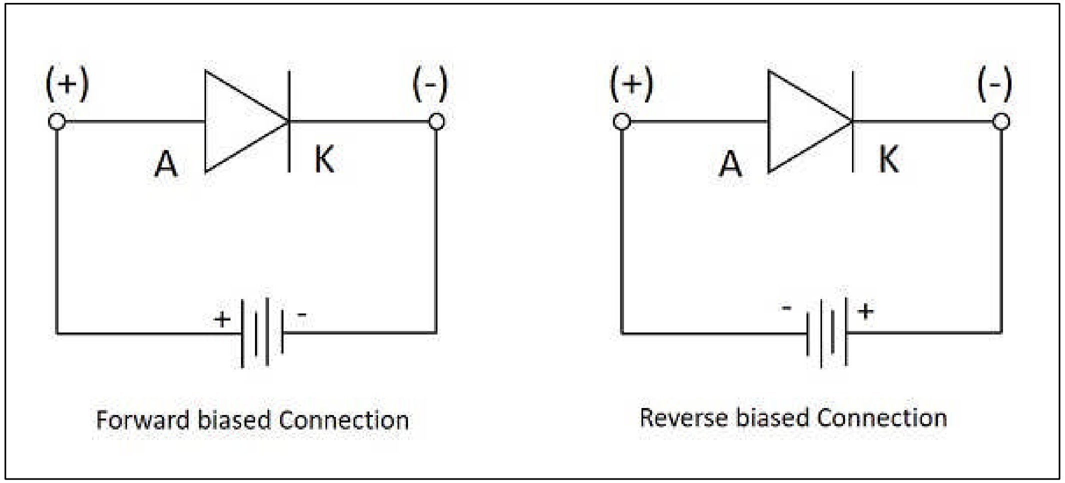


Diode

- A semiconductor diode is a two terminal electronic component with a PN junction. This is also called as a Rectifier.

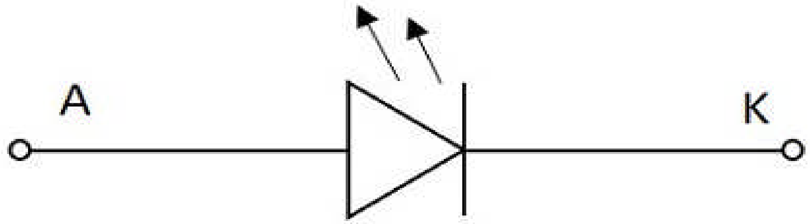
- Biasing of Diode
  - Forward Bias condition
  - Reverse Bias Condition





LED – Light Emitting Diode

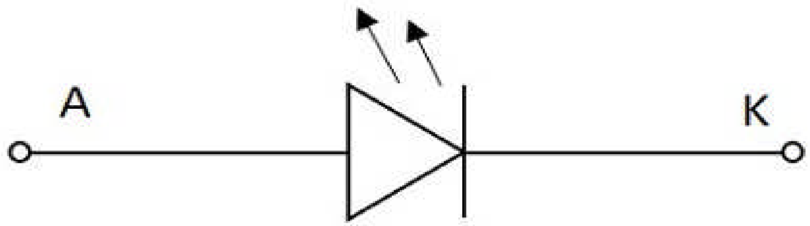
- Like a normal PN junction diode, this is connected in forward bias condition so that the diode conducts. The conduction takes place in a LED when the free electrons in the conduction band combine with the holes in the valence band. This process of recombination **emits light**. This process is called as **Electroluminescence**



Symbol of LED

LED – Light Emitting Diode

- Applications
  - TV Backlighting
  - Smartphone Backlighting
  - LED displays
  - Automotive Lighting
  - Dimming of lights
  - Consumer LED products
  - Many more….

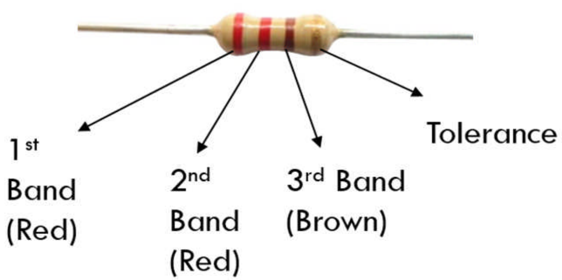


Symbol of LED

Resistor

- Resistor is an electrical component that reduces the electric current. The resistor's ability to reduce the current is called resistance and is measured in units of ohms (symbol: Ω). The resistor's resistance limits the flow of electrons through a circuit.
- How to Calculate the value of Resistor?

3 Band Resistor Resistance Calculation



Color	Color	1st Band	2nd Band	3rd Band Multiplier	4th Band Tolerance
Black		0	0	x1Ω	
Brown		1	1	x10Ω	±1%
Red		2	2	x100Ω	±2%
Orange		3	3	x1kΩ	
Yellow		4	4	x10kΩ	
Green		5	5	x100kΩ	±0.5%
Blue		6	6	x1MΩ	±0.25%
Violet		7	7	x10MΩ	±0.10%
Grey		8	8	x100MΩ	±0.05%
White		9	9	x1GΩ	
Gold				x0.1Ω	±5%
Silver				x0.01Ω	±10%

Ohm’s Law

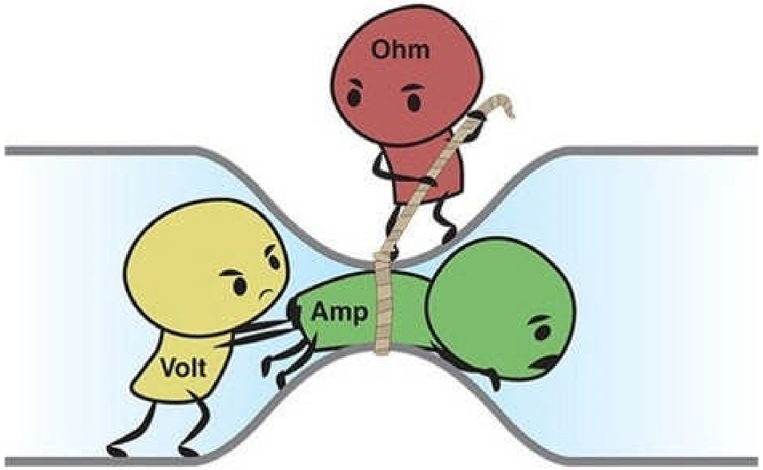
- Ohm’s law states that the current through a conductor between two points is directly proportional to the voltage or potential difference between the two points provided the temperature is constant for a constant length and area.
- Voltage= Current× Resistance

V= I×R

Where, V= voltage (Unit: volts or V)

I= current (Unit: Amperes or A)

R= resistance (Unit: ohms or Ω)



Source: https://sites.google.com/view/trainingday-1/basics-of-electronics/components?authuser=0

Buzzer

- There are many ways to communicate between the user and a product. One of the best ways is audio communication using a buzzer
- **What is buzzer?**
- An audio signalling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound

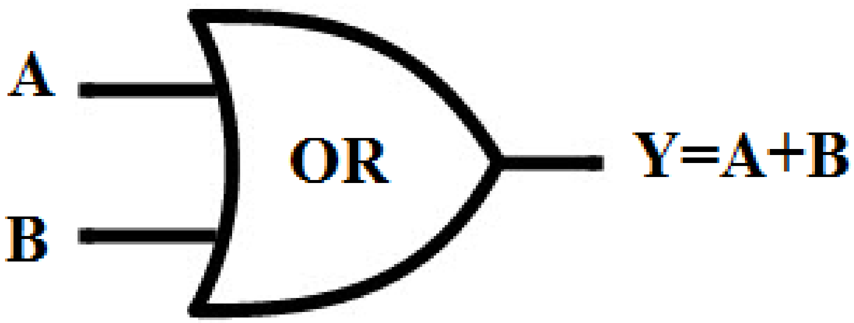
Logic Gates

- A Logic gate is a kind of the basic building block of a digital circuit having two inputs and one output. The input and output relationship are based on a certain logic.
- These gates are implemented using electronic switches such as diodes, transistors. But, in practice, the basic logic gates are built using CMOS technology, MOSFET (Metal Oxide Semiconductor FET), FETS
- The basic logic gates are categorized into seven types as AND, OR, XOR, NAND, NOR, XNOR, and NOT.

OR Gate

Boolean expression of OR gate can be given by,

<u>Y = A + B,</u>



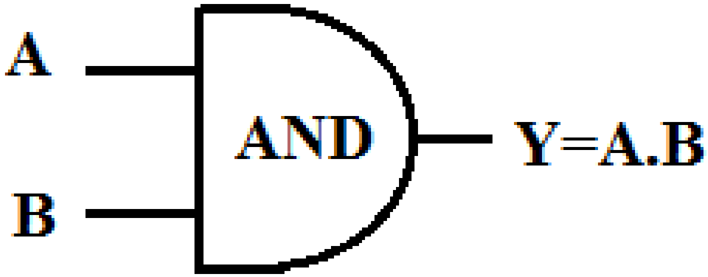
which reads as Y equals A ‘OR’ B.

Inputs		Output
A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

AND Gate

Boolean expression of AND gate can be given by,

Y = A . B,



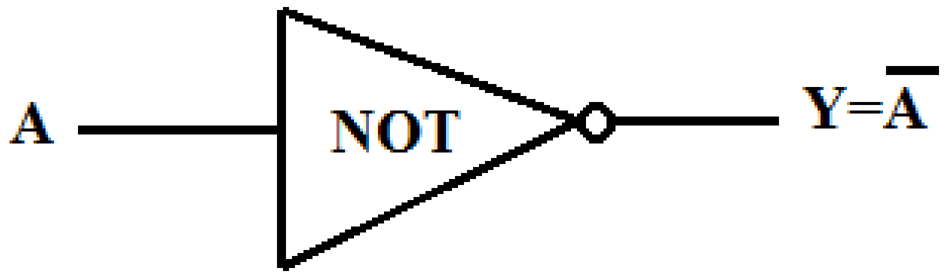
which reads as Y equals A ‘AND’ B.

Inputs		Output
A	B	Y=A.B
0	0	0
0	1	0
1	0	0
1	1	1

NOT Gate

Boolean expression of AND gate can be given by,

$Y = \bar{A}$



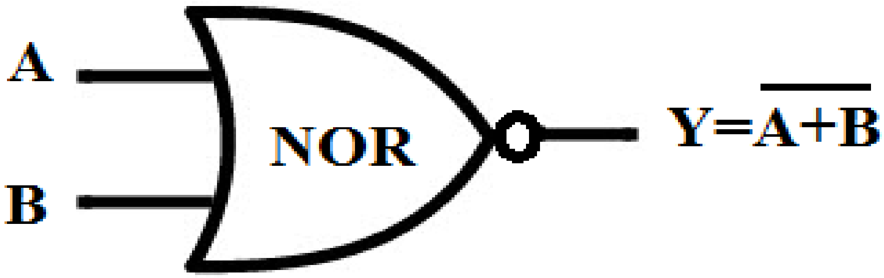
Input	Output
A	Y=$\bar{A}$
0	1
1	0

which reads as Y equals A bar

NOR Gate

Boolean expression of NOR gate can be given by,

<u>Y = A + B (Bar)</u>



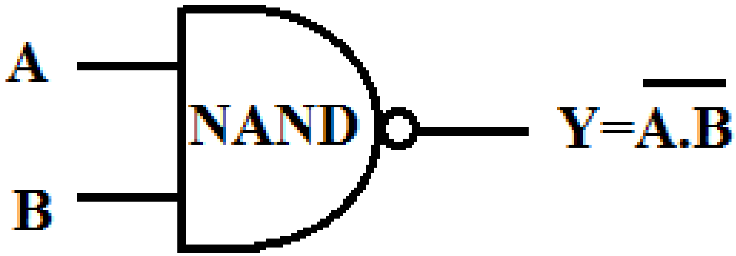
Inputs		Output
A	B	$Y=\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

which reads as Y equals A 'NOR' B

NAND Gate

Boolean expression of NAND gate can be given by,

Y = A . B (Bar)



which reads as Y equals A 'NAND' B

Inputs		Output
A	B	$Y=\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

Logic Gates Summary

- For AND Gate – If both the inputs are high then the output is also high
- For OR Gate – If a minimum of one input is high then the output is High
- For NOT Gate – one input with one inverting output
- NAND Gate – If the minimum one input is low then the output is high
- NOR Gate – If both the inputs are low then the output is high.

Electronic Signal

- A Signal can be understood as "a representation that gives some information about the data present at the source from which it is produced." This is usually time varying.
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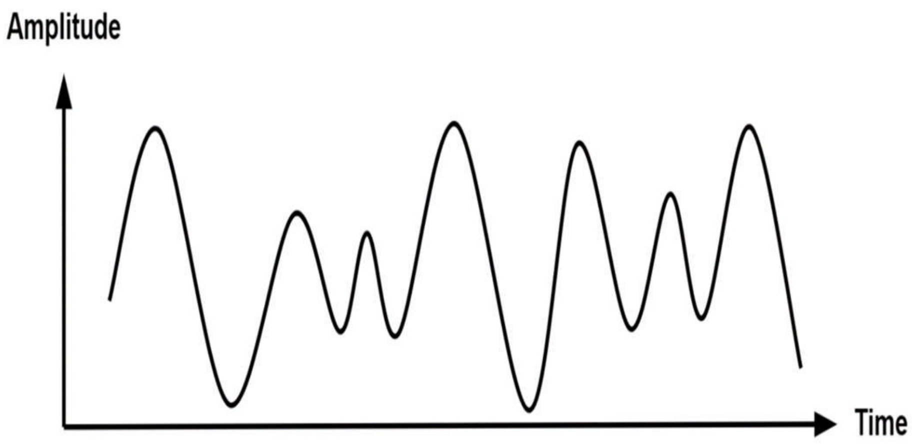
- Signals can be classified either as Analog or Digital, depending upon their characteristics.

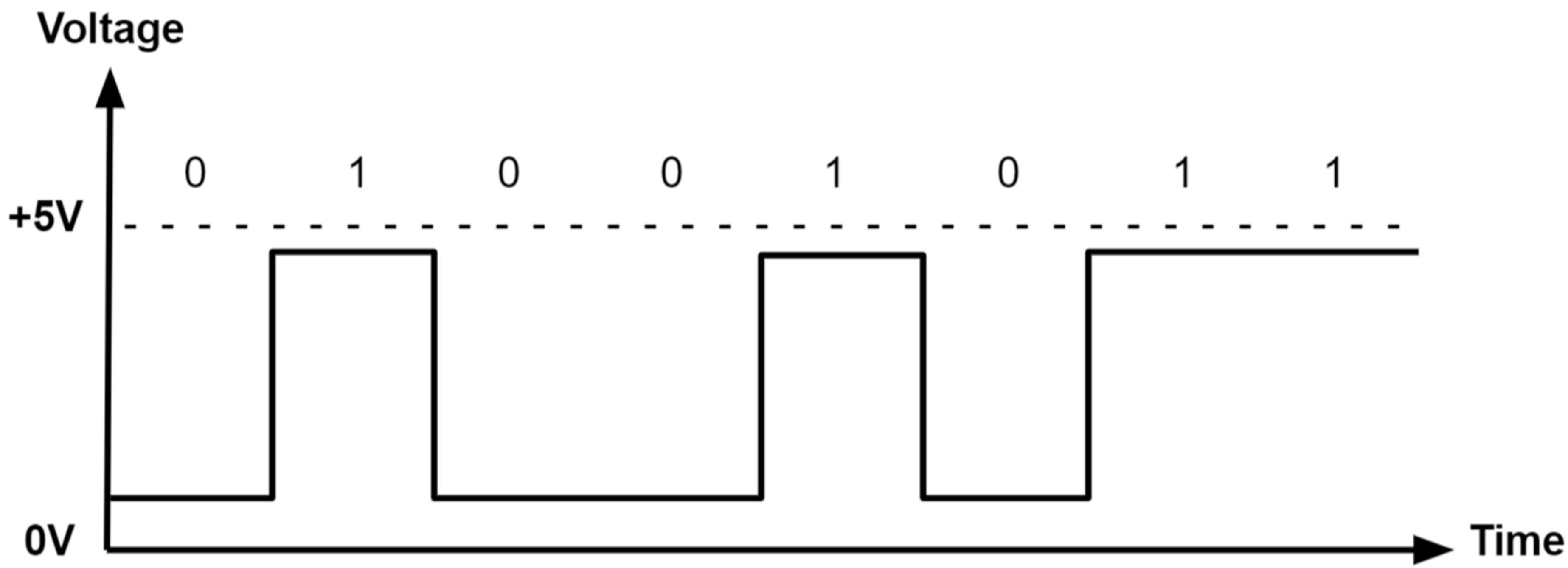
Analog Signal

- An analog signal is time-varying and generally bound to a range (e.g. +12V to -12V), but there is an infinite number of values within that continuous range. An analog signal uses a given property of the medium to convey the signal's information, such as electricity moving through a wire. In an electrical signal, the voltage, current, or frequency of the signal may be varied to represent the information.



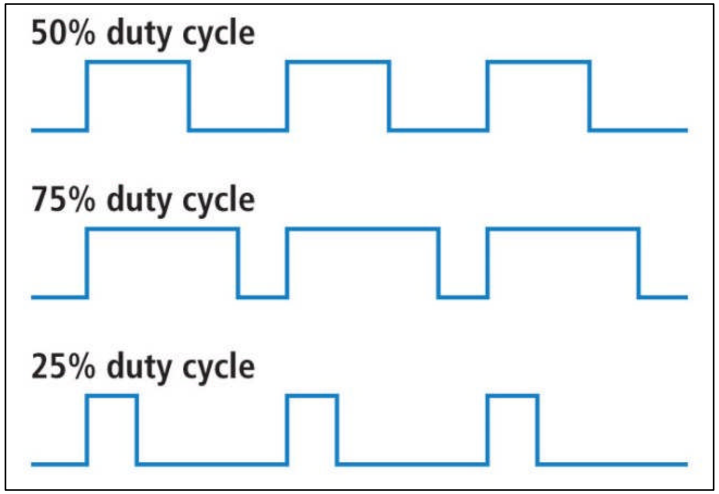
Digital Signal

- A digital signal is a signal that represents data as a sequence of discrete values



Pulse Width Modulation - PWM

- In power electronics, pulse width modulation is a proven effective technique that is used to control semiconductor devices.
- Pulse width modulation or PWM is a commonly used control technique that generates analog signals from digital devices such as microcontrollers.
- The signal thus produced will have a train of pulses, and these pulses will be in the form of square waves.



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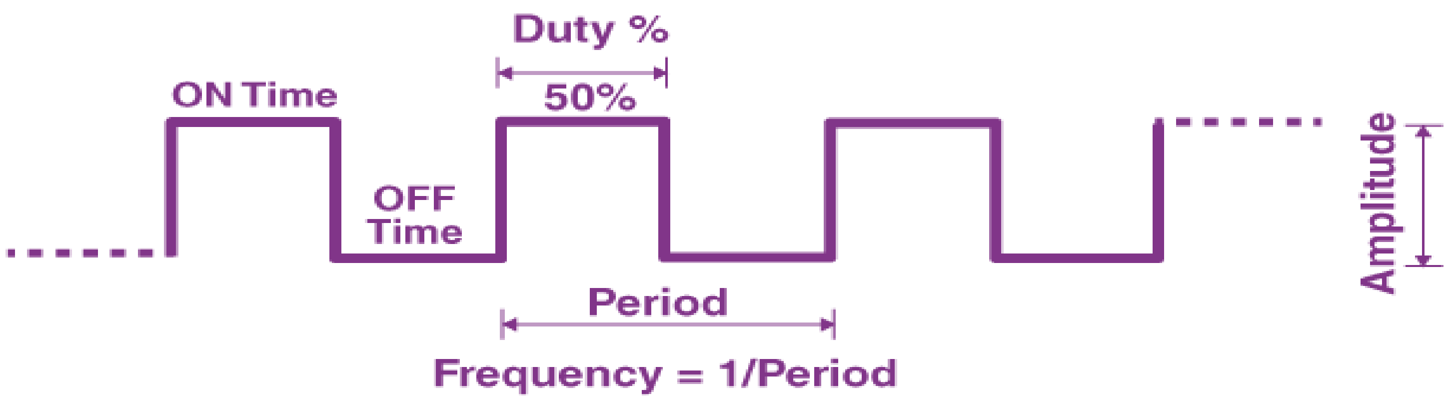
Pulse Width Modulation – Duty Cycle

- PWM signal stays "*ON*" for a given time and stays "*OFF*" for a certain time. The percentage of time for which the signal remains "*ON*" is known as the duty cycle

$$\boxed{\text{Duty Cycle} = \frac{\text{Turn ON time}}{\text{Turn ON time} + \text{Turn OFF time}}}$$

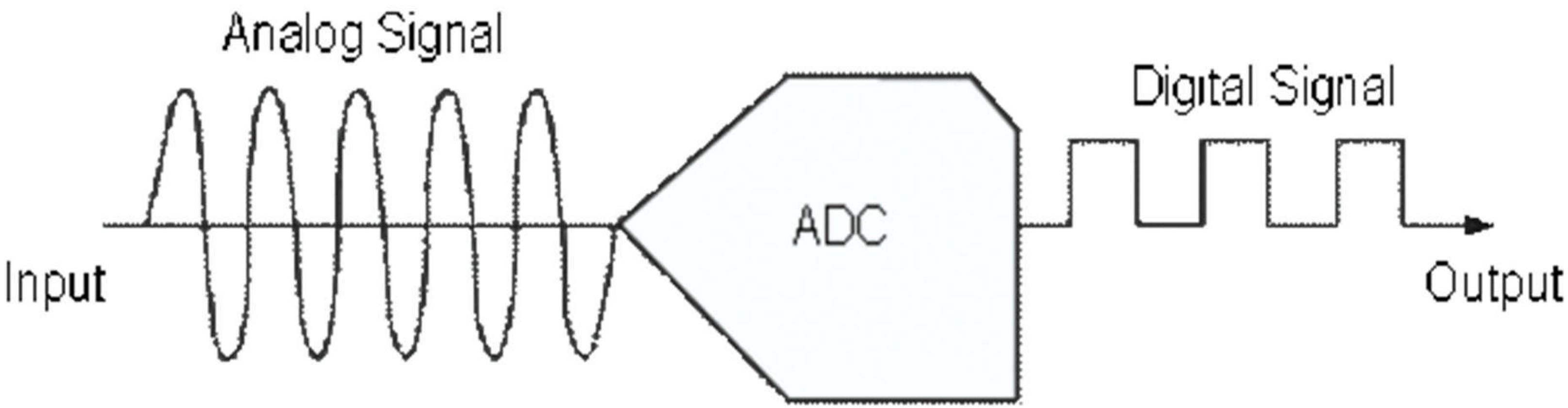


Pulse Width Modulation – Application

- The pulse width modulation technique is used in telecommunication for encoding purposes.
- The PWM helps in voltage regulation and therefore is used to control the speed of motors.
- The PWM technique controls the fan inside a CPU of the computer, thereby successfully dissipating the heat.
- PWM is used in Audio/Video Amplifiers.

Analog to Digital Converter - ADC

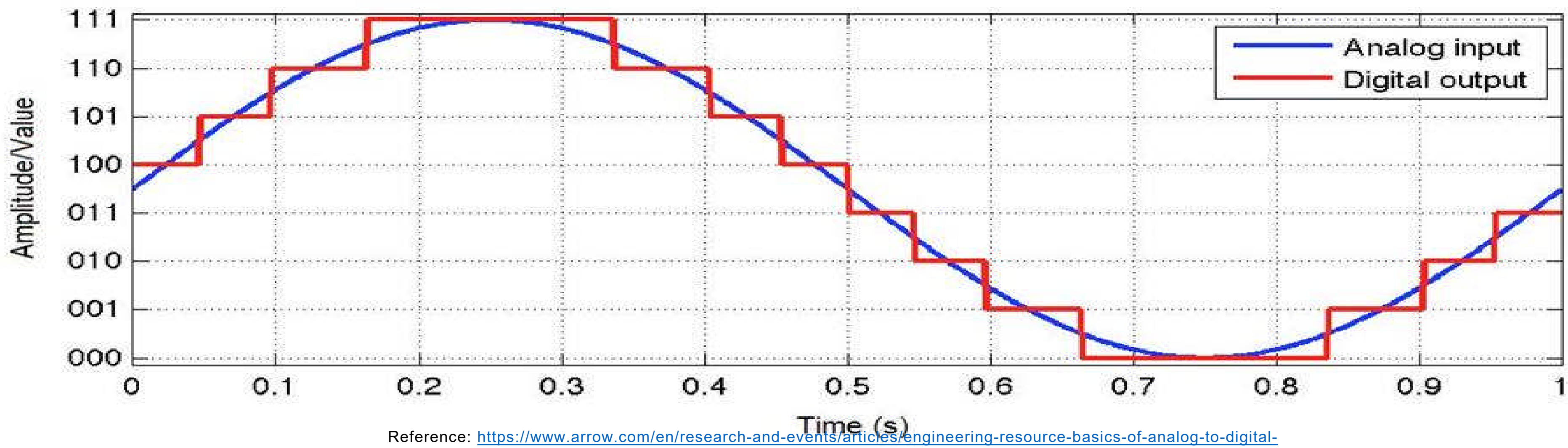
- Analog-to-Digital converters (ADC) translate analog signals, real world signals like temperature, pressure, voltage, current, distance, or light intensity, into a digital representation of that signal



Reference: https://wiki.analog.com/university/courses/electronics/text/chapter-20

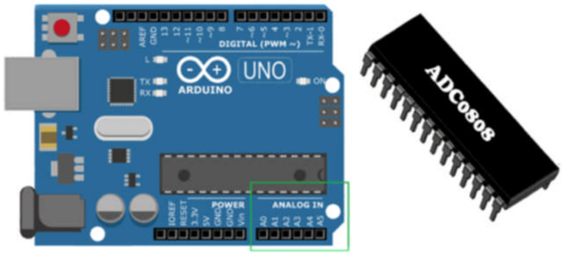
Analog to Digital Converter - ADC

- Digital signals are represented by a sequence of discrete values where the signal is broken down into sequences that depend on the time series or sampling rate. The easiest way to explain this it through a visual! Figure shows a great example of what analog and digital signals look like.



Reference: https://www.arrow.com/en/research-and-events/articles/engineering-resource-basics-of-analog-to-digital-converters#:~:text=ADCs%20follow%20a%20sequence%20when,its%20sampling%20rate%20and%20resolution

Analog to Digital Converter - Application



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